

NERCHINSKIY ZAVOD, Russian Federation

WMO: 308790

Lat: 51.317N

Lon: 119.617E

Elev: 619

StdP: 94.11

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.8	-34.6	-38.8	0.1	-36.5	-36.6	0.1	-34.3	4.9	-18.6	4.2	-19.0	0.0	90	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	29.5	18.2	27.5	17.7	25.7	17.2	20.4	26.0	19.4	24.6	18.5	23.4	2.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	14.5	23.1	17.6	13.6	21.9	16.7	12.8	21.0	62.1	26.2	58.4	24.7	55.2	23.3	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.7	4.9	4.2	-38.5	32.8	3.3	2.3	-40.9	34.4	-42.8	35.7	-44.6	37.0	-46.9	38.7		
			-38.5	22.2	3.1	1.4	-40.8	23.2	-42.6	24.0	-44.4	24.8	-46.7	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.6	-26.3	-20.8	-9.7	1.9	9.9	16.7	19.2	16.7	9.7	0.1	-13.8	-24.2
	DBStd	16.74	4.84	5.44	6.38	4.75	4.01	3.57	2.61	3.25	4.14	5.02	6.65	5.09
	HDD10.0	5036	1127	862	610	245	52	2	0	2	55	308	714	1061
	HDD18.3	7370	1385	1095	868	492	263	70	19	68	261	566	964	1319
	CDD10.0	798	0	0	0	4	49	203	286	210	46	1	0	0
	CDD18.3	90	0	0	0	0	1	22	48	19	1	0	0	0
	CDH23.3	1057	0	0	0	2	47	340	462	188	17	0	0	0
	CDH26.7	277	0	0	0	0	9	100	126	39	2	0	0	0
Wind	WSAvg	1.3	0.5	0.7	1.4	2.0	2.1	1.5	1.3	1.3	1.5	1.5	0.8	0.5
Precipitation	PrecAvg	438	3	4	7	20	27	69	122	99	53	14	9	6
	PrecMax	648	9	10	18	42	76	184	236	220	135	46	26	12
	PrecMin	282	0	0	0	2	4	9	32	23	4	2	1	0
	PrecStd	92	2	3	5	10	17	42	55	46	36	10	6	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.4	-2.6	10.1	21.1	28.0	33.0	32.5	30.5	25.5	17.1	4.7	-9.3
		MCWB	-12.6	-5.3	3.5	9.6	15.1	18.1	19.7	19.6	14.2	8.9	0.8	-10.3
	2%	DB	-15.0	-7.1	5.3	16.9	24.2	29.2	29.7	27.4	22.4	13.0	0.7	-12.4
		MCWB	-15.7	-8.6	0.5	7.7	13.2	17.1	18.9	18.1	13.8	6.5	-2.1	-13.1
	5%	DB	-17.3	-9.9	2.5	13.5	21.4	26.9	27.7	25.3	19.9	10.4	-2.3	-15.0
		MCWB	-17.9	-11.0	-1.4	5.7	11.8	16.6	18.7	17.4	12.5	4.9	-4.2	-15.6
	10%	DB	-19.4	-12.4	-0.1	10.6	18.7	24.8	25.8	23.4	17.4	7.8	-4.5	-17.0
		MCWB	-19.9	-13.3	-2.9	3.9	10.2	16.0	18.1	16.7	11.7	3.5	-5.9	-17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.7	-5.3	4.0	10.9	16.4	20.6	22.5	21.1	16.8	9.1	1.0	-10.3
		MCDB	-11.7	-2.5	9.5	20.3	25.7	28.4	29.6	26.9	20.6	15.8	4.4	-9.4
	2%	WB	-15.7	-8.6	0.8	8.2	14.3	19.1	20.9	19.7	15.0	7.1	-1.8	-13.0
		MCDB	-15.0	-7.0	5.1	16.0	22.5	26.1	26.7	24.7	19.4	12.2	0.5	-12.3
	5%	WB	-17.8	-11.0	-1.2	6.1	12.4	17.8	20.0	18.7	13.6	5.4	-4.1	-15.5
		MCDB	-17.3	-9.9	2.2	13.3	20.1	24.4	24.9	23.5	18.7	9.8	-2.4	-15.1
	10%	WB	-19.9	-13.3	-2.8	4.2	10.9	16.8	19.1	17.6	12.3	3.7	-5.9	-17.4
		MCDB	-19.5	-12.4	-0.2	9.9	17.8	23.0	23.7	21.9	16.9	7.5	-4.5	-17.0
Mean Daily Temperature Range	5% DB	MDBR	8.2	10.7	11.7	11.6	13.0	13.1	11.4	11.5	12.4	11.0	9.8	7.8
		MCDBR	9.9	12.4	13.5	17.2	18.6	17.4	15.5	14.9	16.2	14.9	11.1	8.9
		MCWBR	9.2	11.0	9.3	10.3	9.6	7.5	6.5	6.8	8.1	9.4	8.7	8.2
	5% WB	MCDBR	9.8	12.4	12.9	15.8	16.6	14.5	12.6	12.8	13.6	14.3	10.9	8.8
		MCWBR	9.1	11.1	9.3	10.0	9.1	7.3	6.4	6.7	7.2	9.3	8.5	8.2
Clear-Sky Solar Irradiance	taub	0.249	0.273	0.324	0.421	0.480	0.453	0.419	0.390	0.346	0.317	0.276	0.257	
	taud	2.218	2.183	2.107	2.006	1.925	2.121	2.280	2.334	2.396	2.369	2.275	2.161	
	Ebn at Noon	776	852	863	809	777	807	831	838	840	794	729	684	
	Edh at Noon	68	94	125	157	179	149	125	112	94	79	64	61	
All-Sky Solar Radiation	RadAvg	1.52	2.68	4.26	5.18	5.68	6.00	5.48	4.91	3.83	2.58	1.55	1.17	
	RadStd	0.04	0.09	0.19	0.29	0.45	0.57	0.38	0.41	0.29	0.19	0.07	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	-1.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page